

Prabhav Singh

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RESEARCH STATEMENT

My research interests are focused on method development in NLP, with an emphasis on data-efficient approaches that are adaptable to **supervision constraints** and aligned with how **humans naturally teach, label, and reason**. I am interested in developing novel methods in the *fuzzy area* of model training, where we must learn from partial feedback, conflicting signals, and implicit preferences.

EDUCATION

University of Texas at Austin

PhD - Computer Science

- Research Advisors: Prof. Elias Stengel-Eskin, Prof. Junyi Jessy Li

Texas

Starting Fall 2026

Johns Hopkins University

Masters of Science (Thesis) - Computer Science

- GPA: **3.95/4**, Specialization: Human Language Technologies
- Research Advisors: Prof. Jason Eisner, Prof. Jesus Villalba

Baltimore, MD

May 2026

Delhi University

Bachelor of Engineering - Electrical Engineering

- GPA: **3.83/4**, Field: Instrumentation and Control Engineering
- Research Advisors: Prof. KPS Rana, Prof. Vineet Kumar

Delhi, India

May 2018 - August 2022

PUBLICATIONS

- Sensitivity of Stability: Theoretical & Empirical Analysis of Replicability for Adaptive Data Selection in Transfer Learning:** Prabhav Singh, Jessica Sorrell. *arXiv preprint* 2025, *Under Review at TMLR*
- The JHU-MIT Speaker Recognition System for NIST SRE24: Post-Evaluation Analysis:** Jesus Villalba, J. Borgstrom, Prabhav Singh et al. IEEE Automatic Speech Recognition and Understanding Workshop, ASRU 2025 [*Accepted*]
- Count Your Speakers! Multitask Learning for Multimodal Speaker Diarization:** Prabhav Singh, Jesus Villalba, Najim Dehak. In Proceedings of ISCA Interspeech 2025 [*Oral, Top 20%*].
- EmoJudge: LLM Based Post-Hoc Refinement for Multimodal Speech Emotion Recognition:** Prabhav Singh, Jesus Villalba. In Proceedings of ISCA Interspeech 2025.
- Multimodal Emotion Recognition Harnessing the Complementarity of Speech, Language, and Vision:** Thomas Thebaud, Anna Favaro, Prabhav Singh, et al. In Proc. of the 26th International Conference on Multimodal Interaction (ICMI 2024), ACM [*First Position, EVAC, ICMI'24*].
- SEMI-FND: Stacked Ensemble Based Multimodal Inferencing Framework for Faster Fake News Detection:** Prabhav Singh, Ridam Srivastava, K. Rana, V. Kumar (2023). Expert Systems With Applications, Elsevier, 215, 119302.
- A Topic Modeled Unsupervised Approach to Single Document Extractive Text Summarization:** Ridam Srivastava, Prabhav Singh, K. Rana, V. Kumar (2022). Knowledge Based Systems, Elsevier, 246, 108636.
- A Multimodal Hierarchical Approach To Speech Emotion Recognition From Audio and Text:** Prabhav Singh, Ridam Srivastava, K. Rana, V. Kumar (2021). Knowledge Based Systems, Elsevier, 229, 107316.

PREPRINTS, POSTERS AND PROJECTS

- MARFORMER: A Transformer for Predicting Missing Data Distributions:** Prabhav Singh, Haojun Shi, Xiheng Wang, Jason Eisner (2025). *Under Preparation for TMLR*.
- When LLMs Know They Don't: Probing Latent Representations for Logical Insufficiency:** Matt Wang, Prabhav Singh, Xiheng Wang (2025).
- Active Learning and Feature-Acquisition with LLMs and Humans:** Prabhav Singh, Haojun Shi, Jason Eisner (2025). [*Best Poster Award, MASC-SLL*].
- The JHU-MIT Submission to NIST 2024 Speaker Recognition Evaluation (SRE24):** Jesus Villalba, J. Borgstrom, Prabhav Singh et al. (2024). *Accepted & Presented at NIST SRE24*.

INDUSTRY EXPERIENCE

Machine Learning Engineering Intern

June 2025 – August 2025

Insors.ai

San Jose, CA

- Worked on building multi-agent pipelines to increase zero-touch rate for production APaaS pipelines for 2 clients.
- My project here involved using in-house LLMs to extract structured information from documents and conversations in the accounts payable flow and also building a **Text2SQL** based support agent to allow for structured response to queries.

Founding Machine Learning Engineer

January 2023 – August 2024

Sapper.ai

Bangalore, India

- Responsible for developing **glazeDONUT** - An OCRLess document extraction transformer. The development was an Industry first. **Reduced extraction cost by 65%** per page and **improved accuracy by 30%**.
- Drove the ML efforts for 4 Clients to Production. The projects today contribute to **60% of the firm's revenue**.

Data Scientist - I

July 2022 – January 2023

Ola

Bangalore, India

- Worked on building and engineering data pipelines as part of the EDP team (Data Platform) for **customer facing real-time analytics** for multiple verticals at Ola.
- Migrated Ola's nationwide ride allocation ML Models to a **KubeFlow Platform** to enable MLOps. Managed over **2PB Data migration** from Azure to AWS to **reduce costs by 30%**.

RESEARCH EXPERIENCE

Graduate Research Assistant

August 2024 – Present

Center for Language & Speech Processing, JHU

Baltimore, MD

- Currently working with **Prof. Jason Eisner** at CLSP, JHU. Working on developing ANNOTATIONARENA, a novel approach to make it easier, faster, and cheaper to spin up LLM-based annotation systems and evaluate them.
- Also working on developing the **MARFORMER**, a transformer that imputes missing data by directly predicting the **posterior marginal distribution** of missing distributions using conditional log-likelihood training.
- Also worked with **Prof. Jesus Villalba** on a **novel method** for speaker diarization by **replacing heuristic clustering** in inference stage with learned speaker counts via a multitask multimodal approach.

Undergraduate Research Assistant

October 2018 – April 2022

APC Lab, Delhi University

Delhi, India

- Worked under Prof. KPS Rana and Prof. Vineet Kumar at the APC Lab (NSIT) during my undergraduate degree, where I mostly worked on **multimodal methods in NLP**.
- Published **3 peer-reviewed journal papers** with the lab working on fields like multimodal emotion recognition, benchmarking ensemble techniques for fake news detection and faster approaches for extractive summarization.

SKILLS

Languages: Python, C++, Stan, **Databases:** SQL (Postgres), NoSQL (MongoDB)

ML Areas: Statistical ML, Probabilistic Language Modeling, Language Processing, Speech Processing, Transformers, Active Learning, Reinforcement Learning, ML Learning and Stability Theory

ML Toolkits/Libraries: PyTorch, Tensorflow, HuggingFace, TRL, TTRL, Hyperion, pyAgrum, pgmpy

Infrastructure & Cloud: Apache Flink, Kubernetes, Docker, MLFlow, KubeFlow, AWS, Microsoft Azure

TEACHING AND SERVICES

Reviewer, NeurIPS 2025 (Reliable ML Workshop), ICASSP

Sep - Dec 2025

Graduate CA, Intro to Human Language Technologies (JHU)

Aug – Dec 2025

Graduate CA, NLP for Self-Supervised Learning (JHU)

Jan – May 2025

Instructor (Python), IEEE (Delhi University)

Jan – May 2021

ACTIVITIES AND AWARDS

1. **Premier Research Excellence Award** (2021 & 2022): Awarded by Delhi University for publishing 3 papers in a journal with an impact factor ≥ 8 . Included Semester Scholarship for Spring 2022 and Fall 2021.
2. **Academic Excellence Award** (Fall 2019, Spring 2021): Awarded by Delhi University for ranking top in the 3 of my class during my undergraduate degree.
3. **1st Position (Gridlock 2020)**: Awarded by Flipkart National Hackathon for developing a fashion trend predictor for the E-Commerce Giant in India.
4. **Dramatics and Street Play** (2018 - 2022): Performed as an actor in over 50 street plays as part of Ashwamedh (Dramatics Society of NSIT, DU). Won 7 awards over 4 years across the nation.